THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS FULL SYLLABUS PAPER SET 10

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. $\sqrt{676}$ is equal to: 1

(a) 28 **(b)** 25 **(c)** 27 **(d)** 26

2. How many non-square numbers lie between 11^2 and 12^2? 1

(a) 23 **(b)** 11 **(c)** 21 **(d)** 22

3. One crore is equal to: 1

(a) 10^8 **(b)** 10^7 **(c)** 10^5 **(d)** 10^6

4. 38 in Roman numerals is: 1

(a) XLVIII **(b)** XXXVIII **(c)** XXXIX **(d)** XXXVII

5. Every rectangle is a: 1

(a) rhombus **(b)** parallelogram **(c)** square **(d)** kite

6. Three angles of a quadrilateral are 95°, 105° and 60°. The fourth angle is: 1

(a) 100° **(b)** 80° **(c)** 90° **(d)** 110°

7. The smallest digit that replaces the blank in 52_8 so that the number is divisible by 9 is: 1

(a) 4 **(b)** 5 **(c)** 3 **(d)** 8

8. The value of 103×17 is: 1

(a) 1751 **(b)** 1734 **(c)** 1768 **(d)** 1761

9. If A : B = 2 : 3 and B : C = 4 : 5, then A : B : C is: 1

(a) 2 : 4 : 5 **(b)** 8 : 12 : 15 **(c)** 2 : 3 : 5 **(d)** 6 : 8 : 10

10. A ratio compares two quantities of the same kind by: 1

(a) addition **(b)** subtraction **(c)** division **(d)** multiplication

11. If the price of an item is increased by 10% and then decreased by 10%, the final price is: 1

(a) more than the original **(b)** the same

(c) less than the original **(d)** double

12. The marked price is ₹ 2400 and the discount is 25%. The selling price is: 1

(a) ₹ 600 **(b)** ₹ 3000 **(c)** ₹ 2375 **(d)** ₹ 1800

13. The diagonal of a square of side 10 cm is: 1

(a) $10\sqrt{2}$ cm **(b)** 10 cm **(c)** 20 cm **(d)** $10\sqrt{3}$ cm

14. Is a triangle with sides 10, 11 and 15 a right triangle? 1

(a) Cannot say **(b)** No
(c) Yes **(d)** Only if it is isosceles

15. Three partners invest ₹ 2000, ₹ 3000 and ₹ 5000. The profit ratio is: 1

(a) 1 : 2 : 3 **(b)** 2 : 3 : 5 **(c)** 5 : 3 : 2 **(d)** 2 : 5 : 3

16. A pyramid with a square base has triangular faces numbering: 1

(a) 5 **(b)** 3 **(c)** 4 **(d)** 2

17. The mean of 5 numbers is 32. 4 of them are 25, 30, 35, 40. The remaining number is: 1

(a) 35 **(b)** 28 **(c)** 30 **(d)** 32

18. The product of two numbers with a fixed sum is largest when the numbers are: 1

(a) equal (or as close as possible) **(b)** one is zero
(c) as different as possible **(d)** both odd

19. The area of a trapezium with parallel sides 15 cm and 25 cm and height 8 cm is: 1

(a) 375 cm² **(b)** 160 cm² **(c)** 175 cm² **(d)** 320 cm²

20. If the base of a triangle is doubled and height kept the same, the area: 1

(a) becomes four times **(b)** doubles
(c) halves **(d)** stays the same

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 10–15 words each.

21. Express 0.00045 in scientific notation. 2

22. Convert 1011_2 to decimal. 2

23. What is a palindromic number? Give two examples. 2

24. Divide ₹ 1200 between two friends in the ratio 3 : 5. 2

25. Find the hypotenuse of a right triangle with legs 9 cm and 12 cm. 2

26. Express the scale 1 : 200000 as 1 cm = ? km. 2

27. Why is the median preferred to the mean for incomes? 2

28. Write the formulas for the area of a triangle and a parallelogram. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. Observe the pattern: $1^2 = 1$, $11^2 = 121$, $111^2 = 12321$. Predict 1111^2 and 11111^2. 3

30. Which special quadrilaterals have all of: four equal sides and equal diagonals? 3

31. Show that $(a+b)^2 - (a-b)^2 = 4ab$. 3

32. The price of petrol rises 20%. By what percent must consumption fall to keep the expense the same? 3

33. A cube has edge 5 cm. Find its surface area and the number of such cubes needed to make a cuboid of dimensions $10 \times 5 \times 5$. 3

34. Show that the trick 'Think of n, add 3, square, subtract n^2, subtract 9, divide by n' always gives 6. 3

35. A ladder of length 17 m rests against a wall with its foot 8 m from the wall. How high up the wall does it reach? 3

OR

Using $(n,\ \frac{n^2-1}{2},\ \frac{n^2+1}{2})$ find the triplet for $n = 7$ and verify. 3

36. Find the area of a trapezium with parallel sides 10 cm and 26 cm and distance between them 9 cm. Also find the area of an equal-height rectangle with width 26 cm. 3

OR

A triangular field has base 18 m and height 12 m. Find its area and the cost of levelling at ₹ 12 per m². 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 80–100 words each.

37. **Case study:** A dealer's annual turnover is ₹ 4.8 crore. 5

(i) Write it in standard form.

(ii) If it grows by 10^6 every year, find the turnover after 5 years.

(iii) Write the answer in standard form.

OR

(i) Simplify $\dfrac{16^2 \times 2^3}{4^3}$. 5

(ii) Express the result as a power of 2.

38. (i) Find the principal that amounts to ₹ 5832 in 2 years at 8% CI. 5

(ii) State what is compounding.

OR

(i) How is GST calculated? 5

(ii) A restaurant bill is ₹ 1800 with 5% GST; find the total.

39. (i) Describe how a line graph is read. 5

(ii) Name two real situations best shown by line graphs.

OR

Case study: Marks of two students in 5 tests: A: 70, 72, 74, 76, 78; B: 40, 60, 74, 88, 98. 5

(i) Find both means.

(ii) Which is more consistent?

(iii) Find the ranges.

40. **Case study:** A kite-shaped garden has diagonals 40 m and 30 m perpendicular to each other. 5

(i) Find its area.

(ii) Compare with a rhombus of the same diagonals.

(iii) Cost of grass at ₹ 8 per m².

OR

Case study: A school ground is a trapezium with parallel sides 120 m and 80 m, and the non-parallel sides 50 m each. 5

(i) Find the height.

(ii) Find the area.

(iii) Find the cost of turf at ₹ 15 per m².

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END OF QUESTION PAPER **Total: 80 Marks**

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